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Topic:- Linerization of a Function

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

Question NO: 01

Find linerization of a function $f(x,y) = x^3 + xy + \frac{y^2}{2} + 3$ at point $(2,3)$

Solution:

$f(x,y) = x^3 + xy + \frac{y^2}{2} + 3$

$f(2,3) = (2)^3 + (2)(3) + \frac{(3)^2}{2} + 3$

$f(2,3) = 8 + 6 + \frac{9+6}{2}$

$f(2,3) = \frac{16+12+15}{2} = \frac{43}{2}$

$f_x(x,y) = 3x^2 + y$

$f_x(2,3) = 3(2)^2 + 3 = 15$

$f_y(x,y) = x + y$

$f_y(2,3) = 2+3 = 5$

So,

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

Putting the values, So get

DATE:

$L(x,y$

$L(x,y$

$L(x,y$

Question

$+43y$

Solution

$f(x,y) =$

$f(-1,2)$

$f(-1,2)$

$f(-1,2)$

$f_x(x,y$

$f_x(-1,2$

$f_x(-1,2$

$f_y(x,y$

$f_y(-1,2$

Now,

$L(x,y) =$

Putting

$L(x,y)$

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$L(x,y) = \frac{43}{2} + 15(x-2) + 5(y-3)$

$L(x,y) = \frac{43}{2} + 15x - 30 + 5y - \cancel{15}$ → Yaha cutting nahi hoti

$L(x,y) = 15x + 5y - 45 + \frac{43}{2}$ Ans.

Question: No: 02 ⇒ $f(x,y) = \frac{x^3}{3} + \frac{y}{x} + 43y$ at $(-1,2)$. Find linearization.

Solution:

$f(x,y) = \frac{x^3}{3} + \frac{y}{x} + 43y$

$f(-1,2) = \frac{(-1)^3}{3} + \frac{(2)}{(-1)} + 43(2)$

$f(-1,2) = -\frac{1}{3} - \frac{2}{1} + 86$

$f(-1,2) = \frac{-1-6+258}{3} = \frac{251}{3}$

$f_x(x,y) = \frac{3x^2}{3} - \frac{y}{x^2}$

$f_x(-1,2) = \frac{\cancel{3}(-1)^2}{\cancel{3}} - \frac{2}{(-1)^2} = \cancel{\frac{-3}{3}}$

$f_x(-1,2) = 1 - 2 = -1$

$f_y(x,y) = \frac{1}{x} + 43$

$f_y(-1,2) = \frac{1}{-1} + 43 \Rightarrow -1 + 43 \Rightarrow 42$

Now,

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

Putting the values, so get

$L(x,y) = \frac{251}{3} + (-1)(x-(-1)) + 42(y-2)$

$L(x,y) = \frac{251}{3} - (x+1) + 42y - 84$

$L(x,y) = \frac{251}{3} - x - 1 + 42y - 84$

$L(x,y) = -x + 42y - 85 + \frac{251}{3}$ Ans.

$L(x,y) = -x + 42y \frac{-255+251}{3}$

$L(x,y) = 42y - x - \frac{4}{3}$ Ans.

Question: No: 03 => $f(x,y,z) = xy + yz + xz + \frac{4z}{y}$ at $(4,-2,-1)$. Find Linearization.

Solution:

$f(x,y,z) = xy + yz + xz + \frac{4z}{y}$

$f(4,-2,-1) = (4)(-2) + (-2)(-1) + (4)(-1) + \frac{\cancel{4}^{2}(-1)}{\cancel{-2}_{1}}$

$f(4,-2,-1) = -8 + 2 - 4 + 2 \Rightarrow = -8 + \cancel{4} - \cancel{4}$

$f(4,-2,-1) = -8$

$f_x(x_0,y_0,z_0) = y + z$

$f_x(4,-2,-1) = (-2,-1) = -3$

$f_y(x_0,y_0,z_0) = x + z - \frac{4z}{y^2}$

$f_y(4,-2,-1) = 4 + (-1) - \frac{4(-1)}{(-2)^2}$

$f_y(4,-2,-1) = 4 - 1 + \frac{\cancel{4}}{\cancel{4}}$

$f_y(4,-2,-1) = 4 \cancel{-1} \cancel{+1} \Rightarrow = 4$

$f_z(x_0,y_0,z_0) = y + x + \frac{4}{y}$

$f_z(4,-2,-1) = -2 + 4 + \frac{\cancel{4}^{2}}{\cancel{-2}_{1}} \Rightarrow -2 + 4 - 2$

$f_z(4,-2,-1) = -4 + 4$

$f_z(4,-2,-1) = 0$

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$L(x_0, y_0, z_0)$ ...

Putting

$L(4,-2,-1) =$

$L(4,-2,-1) =$

$L(4,-2,-1)$

* Thomas

Question: N...

at (a) =>

Solution:

(A) => (0,0)

$f(0,0$ ...

$f_x(x_0,y_0) =$

$f_x(0,0) =$

$f_y(x_0,y_0) =$

$f_y(0,0) =$

So,

$L(x,y) = f$ ...

Putt...

$L(x,y) =$

$L(x,y) =$

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$L(x_0, y_0, z_0) = f(x_0, y_0, z_0) + f_x(x_0, y_0, z_0)(x - x_0)$
$+ f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0)$

Putting the values, so get

$L(4, -2, -1) = -8 + (-3)(x-4) + 4(y+2) + 0(z+1)$

$L(4,-2,-1) = -8 - 3x + 12 + 4y + 8 + 0$

$L(4,-2,-1) = 4y - 3x + 12$ Ans.

* Thomas Calculus Question 25-30.

Question: NO: 25 => $f(x, y) = x^2 + y^2 + 1$

at (a) => (0,0), (b) => (1,1)

Solution:

(A) => (0,0) $f(x, y) = x^2 + y^2 + 1$

$f(0,0) = (0)^2 + (0)^2 + 1 = 1$

$f_x(x_0, y_0) = 2x$

$f_x(0,0) = 2(0) = 0$

$f_y(x_0, y_0) = 2y$

$f_y(0,0) = 2(0) = 0$

So,

$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

Putting the values so get

$L(x, y) = 1 + (0)(x - 0) + 0(y - 0)$

$L(x, y) = 1$

(b) => (1,1) $f(x,y) = x^2+y^2+1$

$f(1,1) = (1)^2+(1)^2+1 = 3$

$f_x(x_0,y_0) = 2x$

$f_x(1,1) = 2(1) = 2$

$f_y(x_0,y_0) = 2y$

$f_y(1,1) = 2(1) = 2$

So,

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

Putting the values, so get

$L(x,y) = 3 + 2(x-1) + 2(y-1)$

$L(x,y) = 3+2x-2+2y-2$

$L(x,y) = 2x+2y-1$ Ans.

Question NO: 26 => $f(x,y) = (x+y+2)^2$ at

a. (0,0), b. (1,2).

solution:

(a) (0,0) $f(x,y) = (x+y+2)^2$

$f(0,0) = (0+0+2)^2 = 4$

$f_x(x_0,y_0) = 2(x+y+2).(1)$

$f_x(0,0) = 2(0+0+2) = 4$

$f_y(x_0,y_0) = 2(x+y+2).(1)$

$f_y(0,0) = 2(0+0+2) = 4$

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so,

$$L(x_0,y_0) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$$

Putting the values, so get

$$L(x_0,y_0) = 4 + 4(x-0) + 4(y-0)$$

$$L(x,y) = 4 + 4x - 0 + 4y - 0$$

$$L(x,y) = 4x + 4y + 4 \text{ Ans.}$$

b. $(1,2)$ $f(x,y) = (x+y+2)^2$

$$f(1,2) = (1+2+2)^2 \Rightarrow = (5)^2 = 25$$

$$f_x(x,y) = 2(x+y+2).(1)$$

$$f_x(1,2) = 2(1+2+2) = 10$$

$$f_y(x,y) = 2(x+y+2).(1)$$

$$f_y(1,2) = 2(1+2+2) = 10$$

So,

$$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$$

$$L(x,y) = 25 + 10(x-1) + 10(y-2)$$

$$L(x,y) = 25 + 10x - 10 + 10y - 20$$

$$L(x,y) = 10x + 10y - 30 + 25$$

$$L(x,y) = 10x + 10y - 5 \text{ Ans.}$$

Question No: 27 => $f(x,y) = 3x - 4y + 5$ at

a. $(0,0)$ b. $(1,1)$

Solution:

DATE:

a. $(0,0)$ $\quad f(x,y) = 3x - 4y + 5$

$f(0,0) = 3(0) - 4(0) + 5 = 5$

$f_x(x,y) = 3$

$f_y(x_0,y_0) = -4$

So,

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

$L(x,y) = 5 + 3(x-0) - 4(y-0)$

$L(x,y) = 5 + 3x - 0 - 4y + 0$

$L(x,y) = 3x - 4y + 5$ Ans.

b. $(1,1)$ $\quad f(x,y) = 3x - 4y + 5$

$f(1,1) = 3(1) - 4(1) + 5 = 4$

$f_x(x_0,y_0) = 3$

$f_y(x_0,y_0) = -4$

So,

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

Putting the values so get

$L(x,y) = 4 + 3(x-1) - 4(y-1)$

$L(x,y) = 4 + 3x - 3 - 4y + 4$

$L(x,y) = 3x - 4y + 5$ Ans.

Question: No: 28 => $f(x,y) = x^3y^4$ at

a. $(1,1)$ $\quad$ b. $(0,0)$

Solution:

a. $(1,1)$

$f(1$

$f_x(x,y) =$

$f_x(1,1) =$

$f_y(x,y) =$

$f_y(1,1) =$

So,

$L(x,y) = f$

Putti

$L(x,y) =$

$L(x,y) =$

$L(x,y)$

b. $(0,0)$

$f(0,$

$f_x(x,y)$

$f_x(0,0)$

$f_y(x,y) =$

$f_y(0,0) =$

So,

$L(x,y) = f$

Put

$L(x,y) =$

a. (1,1) $\quad f(x,y) = x^3 y^4$

$f(1,1) = (1)^3(1)^4 = 1$

$f_x(x,y) = 3y^4x^2$

$f_x(1,1) = 3(1)^4(1)^2 = 3$

$f_y(x,y) = 4y^3x^3$

$f_y(1,1) = 4(1)^3(1)^3 = 4$

So,

$l(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

Putting the values, So get

$l(x,y) = 1 + 3(x-1) + 4(y-1)$

$l(x,y) = 1 + 3x - 3 + 4y - 4$

$l(x,y) = 3x + 4y - 6$ Ans.

b. (0,0) $\quad f(x,y) = x^3 y^4$

$f(0,0) = (0)^3(0)^3 = 0$

$f_x(x,y) = 3x^2y^4$

$f_x(0,0) = 3(0)^2(0)^4 = 0$

$f_y(x,y) = 4y^3x^3$

$f_y(0,0) = 4(0)^3(0)^3 = 0$

So,

$l(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

Putting the values. So get

$l(x,y) = 0 + 0(x-0) + 0(y-0) = 0$ Ans.

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<= Question: NO: 29 =>

$f(x,y) = e^x \cos y$ at a. $(0,0)$ b. $(0, \frac{\pi}{2})$

Solution:

a. $(0,0)$ $\quad f(x,y) = e^x \cos y$

$f(0,0) = e^0 \cos 0 = 1.1 = 1$

$f_x(x,y) = e^x \cos y$

$f_x(0,0) = e^0 \cos 0 = 1.1 = 1$

$f_y(x,y) = e^x(-\sin y)$

$f_y(0,0) = -e^0 \sin 0 = -1.0 = 0$

so,

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

Putting the values, so get

$L(x,y) = 1 + 1(x-0) + 0(y-0)$

$L(x,y) = 1 + x$ Ans.

b. $(0, \frac{\pi}{2})$ $\quad f(x,y) = e^x \cos y$

$f(0, \frac{\pi}{2}) = e^0 \cos \frac{\pi}{2} = 1.0 = 0$

$f_x(x,y) = e^x \cos y$

$f_x(0, \frac{\pi}{2}) = e^0 \cos \frac{\pi}{2} = 1.0 = 0$

$f_y(x,y) = e^x(-\sin y)$

$f_y(0, \frac{\pi}{2}) = -e^0 \sin \frac{\pi}{2} = -1.1 = -1$

so,

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

Putting the values, so get

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$b.(0, \frac{\pi}{2})$

$L(x,y) = 0 + 0(x-0) + (-1)(y-\frac{\pi}{2})$

$L(x,y) = -y + \frac{\pi}{2}$ Ans.

$(y-y_0)$

Question: NO: 30 => $f(x,y) = e^{2y-x}$ at

a. $(0,0)$ b. $(1,2)$

solution:

a. $(0,0)$ $f(x,y) = e^{2y-x}$

$f(0,0) = e^{2(0)-(0)} = 1$

$f_x(x_0,y_0) = e^{2y-x}.(-1)$

$f_x(0,0) = -e^{2(0)-(0)} = -1$

$f_y(x_0,y_0) = e^{2y-x}(2)$

$f_y(0,0) = 2e^{2(0)-(0)} = 2(1) = 2$

so,

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

Putting the values so get

$L(x,y) = 1 + (-1)(x-0) + (2)(y-0)$

$L(x,y) = 1 - x + 0 + 2y - 0 = -x + 2y + 1$ Ans.

b. $(1,2)$ $f(x,y) = e^{2y-x}$

$f(1,2) = e^{2(2)-(1)} = e^3$

$f_x(x_0,y_0) = e^{2y-x}.(-1)$

$f_x(1,2) = -e^{2(2)-(1)} = -e^3$

$f_y(x_0,y_0) = e^{2y-x}.(2)$

$f_y(1,2) = 2e^{2(2)-(1)} = 2e^3$

$)(y-y_0)$

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So,

$$L(x,y_0) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$$

Putting the values, so get

$$L(x,y) = e^3 + (-e^3)(x-1) + 2e^3(y-2)$$

$$L(x,y) = \cancel{e^3} \cancel{-e^3} e^3 - e^3x + e^3 + 2e^3y - 4e^3$$

$$L(x,y) = 2e^3y - e^3x - 2e^3$$

$$L(x,y) = e^3(2y - x - 2)$$ Ans.

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Topic:-

Equation of tangent plane And normal line

Question: NO: 01 => Find equation of tangent and normal line to the surface $4x^2 - y^2 + 3z^2 - 10$ at $P(2,3,1)$.

Solution:

$$f(x,y,z) = 4x^2 - y^2 + 3z^2 - 10$$

$$\nabla f = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j + \frac{\partial f}{\partial z} k$$

$$\nabla f = 8x^2 i - 2y j + 6zk$$

$$\nabla f \Big|_{(2,-3,1)} = 8(2)^2 i + 2(3) j + 6(1) k$$

$$= 16 i + 6 j + 6k$$

i) Equation of tangent plane

$$(x - x_1)(\nabla f)_x + (y - y_1)(\nabla f)_y + (z - z_1)(\nabla f)_z = 0$$

Putting the values, So get

$$(x-2)(16) + (y+3)(6) + (z-1)6 = 0$$

$$16x - 32 + 6y + 18 + 6z - 6 = 0$$

$$2(8x - 16 + 3y + 9 + 3z - 3) = 0 \rightarrow \text{taking common } 2$$

$$8x + 3y + 3z - 19 + 9 = \frac{0}{2}$$

$$8x + 3y + 3z = 10 \text{ Ans.}$$

ii) Equation of normal line

$$\frac{x - x_1}{(\nabla f)_x} = \frac{y - y_1}{(\nabla f)_y} = \frac{z - z_1}{(\nabla f)_z}$$

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$$\frac{x-2}{16} = \frac{y+3}{6} = \frac{z-1}{6}$$

multiply each by 2, so get

$$2 \times \frac{x-2}{16} = 2 \times \frac{y+3}{6} = 2 \times \frac{z-1}{6}$$

$$\frac{x-2}{8} = \frac{y+3}{3} = \frac{z-1}{3} \Rightarrow \text{Ans}$$

Question: NO: 02 ⇒ Find equation of tangent and normal line to the surface $9x^2+4y^2-z^2-36=0$ at $P(2,3,6)$.

Solution:

$$f(x,y,z) = 9x^2+4y^2-z^2-36$$

$$\nabla f = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j + \frac{\partial f}{\partial z} k$$

$$\nabla f = 18x i + 8y j - 2z k$$

$$\nabla f\Big|_{(2,3,6)} = 18(2) i + 8(3) j - 2(6) k$$

$$= 36 i + 24 j - 12 k$$

(i) Equation of tangent plane:

$$(x-x_1)(\nabla f)_x + (y-y_1)(\nabla f)_y + (z-z_1)(\nabla f)_z = 0$$

Putting the values so get

$$(x-2)(36) + (y-3)(24) + (z-6)(-12) = 0$$

$$36x - 72 + 24y - \cancel{72} - 12z + \cancel{72} = 0$$

$$36x + 24y - 12z - 72 = 0 \rightarrow \text{taking 12 as common}$$

$$12(3x+2y-z-6) = 0$$

$$3x+2y-z-6 = \frac{0}{2}$$

$$3x+2y-z = 6$$

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(ii) Equation of normal line

$$\frac{(x-x_1)}{(\nabla f)_x} = \frac{(y-y_1)}{(\nabla f)_y} = \frac{(z-z_1)}{(\nabla f)_z}$$

Putting the values so get

$$\frac{x-2}{36} = \frac{y-3}{24} = \frac{z-6}{-12}$$

multiply by 12, so get

$$12\times\frac{x-2}{36_3} = 12\times\frac{y-3}{24_2} = 12\times\frac{z-6}{-12_1}$$

$$\frac{x-2}{3} = \frac{y-3}{2} = -(z-6) \Rightarrow \text{Ans.}$$

Book Questions

Question: NO: 20 => Find an equation of the tangent plane at any point $P(x_1, y_1, z_1)$ of the elliptic paraboloid $z = x^2 + 4y^2$.

Solution:

$$f(x,y,z) = x^2 + 4y^2 - z$$

$$\nabla f = \frac{\partial f}{\partial x}\underline{i} + \frac{\partial f}{\partial y}\underline{j} + \frac{\partial f}{\partial z}k$$

$$\nabla f = 2x\underline{i} + 8y\underline{j} - 1k$$

$$\nabla f\Big|_{(x_1,y_1,z_1)} = 2x_1\underline{i} + 2y_1\underline{j} + z_1 k$$

(i) Equation of tangent plane:

$$(x-x_1)(\nabla f)_x + (y-y_1)(\nabla f)_y + (z-z_1)(\nabla f)_z = 0$$

Putting the values so get

of

(2,3,6).

= 0

0

as common

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$(x-x_1)(2x_1)+(y-y_1)(8y_1)+(z-z_1)(-z_1)=0$

$2xx_1-2x_1^2+8yy_1-8y_1^2-zz_1+z_1^2=0$ Ans

Question: NO: 11 => Find the equation of the tangent plane and the normal line to the surface $x^2+y^2+z^2=14$ at $P(1,-2,3)$.

Solution:

$x^2+y^2+z^2=14 \Rightarrow x^2+y^2+z^2-14=0$

$f(x,y,z)=x^2+y^2+z^2-14$

$\nabla f=\frac{\partial f}{\partial x}\underline{i}+\frac{\partial f}{\partial y}\underline{j}+\frac{\partial f}{\partial z}\underline{k}$

$\nabla f=2x\underline{i}+2y\underline{j}+2z\underline{k}$

$\nabla f\big|_{(1,-2,3)}=2(1)\underline{i}+2(-2)\underline{j}+2(3)\underline{k}$

$=2\underline{i}-4\underline{j}+6\underline{k}$

(i) Equation of tangent plane:

~~Solution~~: $(x-x_1)(\nabla f)_x+(y-y_1)(\nabla f)_y+(z-z_1)(\nabla f)_z=0$

Putting the values, so get

$(x-1)(2)+(y+2)(-4)+(z-3)(6)=0$

$2x-2-4y-8+6z-18=0$

$6z+2x-4y-28=0$

$\frac{2x}{2}-\frac{4y}{2}+\frac{6z}{2}=\frac{28}{2}$ Ans. divided by 2, so

$x-2y+3z=14$ Ans.

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(ii) Equation of normal line:

$$\frac{(x-x_1)}{(\nabla f)_x} = \frac{(y-y_1)}{(\nabla f)_y} = \frac{(z-z_1)}{(\nabla f)_z}$$

Putting the values, so get

$$\frac{x-1}{2} = \frac{y+2}{-4} = \frac{z-3}{6}$$

multiply by 2, so get

$$2\times\frac{x-1}{2} = 2\times\frac{y+2}{-4_2} = 2\times\frac{z-3}{6_3}$$

$$x-1 = -\frac{y+2}{2} = \frac{z-3}{3} \text{ Ans.}$$

Question: NO: 12 => Find the equation of the tangent plane and the normal line to the surface $x^2-2y^2-z^2=4$ at $P(-6, 2, \sqrt{24})$.

Solution:

$$x^2-2y^2-z^2=4 \Rightarrow x^2-2y^2-z^2-4=0$$

$$f(x,y,z) = x^2-2y^2-z^2-4$$

$$\nabla f = \frac{\partial f}{\partial x}\underline{i} + \frac{\partial f}{\partial y}\underline{j} + \frac{\partial f}{\partial z}\underline{k}$$

$$\nabla f = 2x - 4y - 2z$$

$$\nabla f\Big|_{(-6,2,\sqrt{24})} = 2(-6)\underline{i} - 4(2)\underline{j} - 2(\sqrt{24})\underline{k}$$

$$\nabla f = -12\underline{i} - 8\underline{j} - 2\sqrt{24}\,\underline{k}$$

(i) Equation of tangent plane:

$$(x-x_1)(\nabla f)_x + (y-y_1)(\nabla f)_y + (z-z_1)(\nabla f)_z = 0$$

Putting the values so get

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$[\{x-(-6)\}(-12)] + (y-2)(-8) + (z-\sqrt{24})(-2\sqrt{24}) = 0$

$-12x - 72 - 8y + 16 - 2\sqrt{24}z + 48 = 0$

$-12x - 8y - 2\sqrt{24}z - 8 = 0$

$-12x - 8y - 2\sqrt{24}z = 8 \Rightarrow$ Ans.

(ii) Equation of simple line:

Solution $\frac{x-x_1}{(\nabla f)_x} = \frac{y-y_1}{(\nabla f)_y} = \frac{z-z_1}{(\nabla f)_z}$

Putting the values, so get

$$\frac{x+6}{-12} = \frac{y-2}{-8} = \frac{z-\sqrt{24}}{-2\sqrt{24}}$$

multiply by 4, so get

$$4 \times \frac{x+6}{-12} = 4 \times \frac{y-2}{-8} = 4 \times \frac{z-\sqrt{24}}{-2\sqrt{24}\sqrt{6}}$$

$$-\frac{x+6}{3} = -\frac{y-2}{2} = -\frac{z-\sqrt{24}}{\sqrt{6}} \Rightarrow \text{Ans.}$$

Question: NO: 13 => Find the equation of the tangent plane and the normal line to the surface $x^2 + z^2 = \frac{a^2}{h^2}y^2$ at $P\left(\frac{a}{\sqrt{2}}, h, \frac{a}{\sqrt{2}}\right)$

Solution:

$$x^2 + z^2 = \frac{a^2}{h^2}y^2 \Rightarrow x^2 - \frac{a^2}{h^2}y^2 + z^2 = 0$$

$f(x,y,z) = x^2 - \frac{a^2}{h^2}y^2 + z^2$

$\nabla f = \frac{\partial f}{\partial x}\underline{i} + \frac{\partial f}{\partial y}\underline{j} + \frac{\partial f}{\partial z}\underline{k}$

$\nabla f = 2x\underline{i} - 2y\frac{a^2}{h^2}\underline{j} + 2z\underline{k}$

$\nabla f\Big|_{\left(\frac{a}{\sqrt{2}}, h, \frac{a}{\sqrt{2}}\right)} = 2\left(\frac{a}{\sqrt{2}}\right)\underline{i} - 2(h)\left(\frac{a^2}{h^2}\right)\underline{j} + 2\left(\frac{a}{\sqrt{2}}\right)k$

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$$\nabla f = \frac{2a}{\sqrt{2}}\hat{i} - 2\frac{a^2}{h}\hat{j} + \frac{2a}{\sqrt{2}}\hat{k}$$

$$\nabla f = \sqrt{2}a\hat{i} - \frac{2a^2}{h}\hat{j} + \sqrt{2}a\hat{k}$$

i) Equation of tangent line:

$$(x-x_1)(\nabla f)_x + (y-y_1)(\nabla f)_y + (z-z_1)(\nabla f)_z = 0$$

Putting the values, so get

$$\left(x-\frac{a}{\sqrt{2}}\right)(\sqrt{2}a) - \left(2\frac{a^2}{h}\right)(y-h) + \left(z-\frac{a}{\sqrt{2}}\right)(\sqrt{2}a) = 0$$

$$\sqrt{2}ax - \frac{a}{\sqrt{2}}\cdot\sqrt{2}a - 2\frac{a^2}{h}y + 2\frac{a^2}{h}\cdot h + \sqrt{2}az - \frac{a}{\sqrt{2}}\cdot\sqrt{2}a = 0$$

$$\sqrt{2}ax - a^2 - \frac{2a^2}{h}y + 2a^2 + 2\sqrt{a}z - a^2 = 0$$

$$\sqrt{2}ax - \cancel{2a^2} - 2\frac{a^2}{h}y + \cancel{2a^2} + 2\sqrt{a}z = 0$$

$$\sqrt{2}ax - 2\frac{a^2}{h}y + 2\sqrt{a}z = 0 \text{ Ans.}$$

ii) Equation of normal line:

$$\frac{x-x_1}{(\nabla f)_x} = \frac{y-y_1}{(\nabla f)_y} = \frac{z-z_1}{(\nabla f)_z}$$

Putting the values, so get

$$\frac{x-\frac{a}{\sqrt{2}}}{\sqrt{2}a} = \frac{y-h}{2\frac{a^2}{h}} = \frac{z-\frac{a}{\sqrt{2}}}{\sqrt{2}a} \Rightarrow \text{Ans.}$$

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Question No: 14 => Find the equation of the tangent plane and the normal line to the surface $z = e^x \cos y$ at $P(0, \pi/2, 0)$.

Solution:

$z = e^x \cos y \Rightarrow e^x \cos y - z = 0$

$f(x,y,z) = e^x \cos y - z$

$\nabla f = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j + \frac{\partial f}{\partial z} k$

$\nabla f = \cos y \, e^x i - e^x \sin y \, j - 1k$

$\nabla f \Big|_{(0, \pi/2, 0)} = \cos\left(\frac{\pi}{2}\right) e^0 i - e^0 \sin\left(\frac{\pi}{2}\right) j - 1k$

$\nabla f = 1.0 - 1.(+1) - 1k$

$\nabla f = 0i - 1j - 1k$

(i) Equation of tangent line:

$(x - x_1)(\nabla f)_x + (y - y_1)(\nabla f)_y + (z - z_1)(\nabla f)_z = 0$

Putting the values, so get

$(x-0)(0) + \left(y - \frac{\pi}{2}\right)(-1) + (z-0)(-1) = 0$

$0 - y + \frac{\pi}{2} - z = 0$

$-\left(y - \frac{\pi}{2} + z\right) = 0$

$y - \frac{\pi}{2} + z = 0$ Ans.

(ii) Equation of normal line:

$\frac{x - x_0}{(\nabla f)_x} = \frac{y - y_0}{(\nabla f)_y} = \frac{z - z_0}{(\nabla f)_z}$

Putting the values, so get

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$$\frac{x-0}{0} = \frac{y-\pi/2}{-1} = \frac{z-0}{-1} \Rightarrow \text{Ans.}$$

Question: NO: 15 ⇒ Find the equation of the tangent plane and the normal line to the surface $x = \ln\left(\frac{y}{2z}\right)$ at $P(0,2,1)$.

Solution:

$$x = \ln\frac{y}{2z} \Rightarrow +x - \ln\left(\frac{y}{2z}\right) = 0$$

$$f(x,y,z) = -\ln\left(\frac{y}{2z}\right) + x$$

$$\nabla f = \frac{\partial f}{\partial x}\underline{i} + \frac{\partial f}{\partial y}\underline{j} + \frac{\partial f}{\partial z}\underline{k}$$

$$\nabla f = +1\underline{i} + \left(-\frac{1}{y}\right)\underline{j} + \left(\frac{1}{z}\right)\underline{k}$$

$$\nabla f = 1\underline{i} - \frac{1}{y}\underline{j} + \frac{1}{z}\underline{k}$$

$$\nabla f\Big|_{(0,2,1)} = 1\underline{i} - \frac{1}{2}\underline{j} + \frac{1}{1}\underline{k}$$

$$= 1\underline{i} - \frac{1}{2}\underline{j} + 1\underline{k}$$

i) **Equation of tangent line:**

$$(x-x_0)(\nabla f)_x + (y-y_0)(\nabla f)_y + (z-z_0)(\nabla f)_z = 0$$

Putting the values, so get

$$(x-0)(1) + (y-2)\left(-\frac{1}{2}\right) + (z-1)(1) = 0$$

$$x - \frac{1}{2}y + \cancel{1} + z - \cancel{1} = 0$$

$$x - \frac{1}{2}y + z = 0 \Rightarrow \text{Ans.}$$

ii) **Equation of normal line:**

$$\frac{x-x_0}{(\nabla f)_x} = \frac{y-y_0}{(\nabla f)_y} = \frac{z-z_0}{(\nabla f)_z}$$

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Putting the values, So get

$$\frac{x-0}{1} = \frac{y-2}{-1/2} = \frac{z-1}{1}$$

$$x = -2(y-2) = z-1 \Rightarrow \text{Ans.}$$

Question: NO: 16 => Find the equation of the tangent plane and the normal line to the surface $xz = 4$ at $P(-2,2,-2)$.

Solution:

$$xz = 4 \Rightarrow xz - 4 = 0$$

$$f(x,y,z) = xz - 4$$

$$\nabla f = \frac{\partial f}{\partial x}\underline{i} + \frac{\partial f}{\partial y}\underline{j} + \frac{\partial f}{\partial z}\underline{k}$$

$$\nabla f = z\underline{i} + 0\underline{j} + x\underline{k}$$

$$\nabla f\Big|_{(-2,2,2)} = -2\underline{i} + 0\underline{j} - 2\underline{k}$$

(i) Equation of tangent line:

$$(x-x_1)(\nabla f)_x + (y-y_1)(\nabla f)_y + (z-z_1)(\nabla f)_z = 0$$

Putting the values, So get

$$(x+2)(-2) + (y-2)(0) + (z+2)(-2) = 0$$

$$-2x - 4 + 0 - 0 - 2z - 4 = 0$$

$$-2x - 4 - 2z - 4 = 0$$

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$$-2x-2z-8=0$$
$$-2(x+z+4)=0$$
$$x+z+4=\frac{0}{2}$$
$$x+z=-4 \Rightarrow \text{Ans.}$$

(ii) Equation of normal line:

$$\frac{x-x_1}{(\nabla f)_x}=\frac{y-y_1}{(\nabla f)_y}=\frac{z-z_1}{(\nabla f)_z}$$

Putting the values, so get

$$\frac{x+2}{-2}=\frac{y-2}{0}=\frac{z+2}{-2} \Rightarrow \text{Ans.}$$

Question: NO: 17 ⇒ Find the equation of the tangent plane and the normal line to the surface $x^{2/3}+y^{2/3}+z^{2/3}=9$ at $P(1,8,-8)$.

Solution:

$$x^{2/3}+y^{2/3}+z^{2/3}-9=0$$
$$f(x,y,z)=x^{2/3}+y^{2/3}+z^{2/3}-9$$
$$\nabla f=\frac{\partial f}{\partial x}\underline{i}+\frac{\partial f}{\partial y}\underline{j}+\frac{\partial f}{\partial z}\underline{k}$$
$$\nabla f=\frac{2}{3}x^{-1/3}\underline{i}+\frac{2}{3}y^{-1/3}\underline{j}+\frac{2}{3}z^{-1/3}\underline{k}$$
$$\nabla f=\frac{2}{3x^{1/3}}\underline{i}+\frac{2}{3y^{1/3}}\underline{j}+\frac{2}{3z^{1/3}}\underline{k}$$
$$\nabla f\Big|_{(1,8,-8)}=\frac{2}{3(1)}\underline{i}+\frac{2}{3(8)^{1/3}}\underline{j}+\frac{2}{3(8)^{1/3}}\underline{k}$$

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$$\nabla f = \frac{2}{3}\underline{i} + \frac{x}{3(2)^{\frac{2+1}{3}}}\underline{j} - \frac{x}{3(2)^{\frac{2+1}{3}}}\underline{k}$$

$$\nabla f = \frac{2}{3}\underline{i} + \frac{1}{3}\underline{j} - \frac{1}{3}\underline{k}$$

(i) Equation of tangent line:

$$(x-x_1)(\nabla f)_x + (y-y_1)(\nabla f)_y + (z-z_1)(\nabla f)_z = 0$$

Putting the values so get

$$(x-1)\left(\frac{2}{3}\right) + (y-8)\left(\frac{1}{3}\right) + (z+8)\left(-\frac{1}{3}\right) = 0$$

$$\frac{2}{3}x - \frac{2}{3} + \frac{y}{3} - \frac{8}{3} - \frac{z}{3} - \frac{8}{3} = 0$$

$$\frac{2}{3}x + \frac{y}{3} - \frac{z}{3} - \frac{2}{3} - \frac{8}{3} - \frac{8}{3} = 0$$

$$\frac{2x + y - z - 2 - 8 - 8}{3} = 0$$

$$2x + y - z - 18 = 0 \times 3$$

$$2x + y - z - 18 = 0$$

$$2x + y - z = 18 \Rightarrow \text{Ans.}$$

(ii) Equation of normal line:

$$\frac{x-x_1}{(\nabla f)_x} = \frac{y-y_1}{(\nabla f)_y} = \frac{z-z_1}{(\nabla f)_z}$$

Putting the values, so get

$$\frac{x-1}{2/3} = \frac{y-8}{1/3} = \frac{z+8}{-1/3}$$

$$\frac{3(x-1)}{2} = 3(y-8) = -3(z+8) \Rightarrow \text{Ans.}$$